

Algebra & Polynomials

Like Terms · Substitution · Polynomials · Monomial Products & Quotients · Exponent Laws

MTH1W · Homework — Independent Practice

Name: _____

Date: _____

Goal: Practise simplifying, evaluating and operating on polynomials independently.

Show your full working in the space provided under each question. Levels: L1 fluency, L2 application, L3 thinking. Full worked solutions are in the Answer Key.

Level 1 — Fluency

1. Simplify: $7x + 2x - 3x$. **[Like terms]**

2. Evaluate $5x - 4$ when $x = 3$. **[Substitution]**

3. Simplify: $(3a + 2) + (4a - 6)$. **[Add polynomials]**

4. Expand: $6(2x - 3)$. **[Multiply by constant]**

5. Simplify: $(20x + 8) \div 4$. **[Divide by constant]**

6. Simplify: $x^6 \cdot x^3$. **[Exponent law]**

Level 2 — Application

7. Simplify: $5x - 3y - 2x + 8y - 4$. **[Like terms]**

8. Evaluate $3a - 2b$ when $a = 4$ and $b = 5$. **[Substitution]**

9. Simplify: $(6x^2 + 2x - 5) - (2x^2 - 3x + 1)$. **[Subtract polynomials]**

10. Expand: $4x(3x - 2)$. **[Multiply by monomial]**

11. Simplify: $(18x^2 - 12x) \div 6x$. **[Divide by monomial]**

12. Simplify: $(3x^2)(5x^4)$. **[Exponent laws]**

Level 3 — Thinking

13. Priya wrote $2x + 3x = 5x^2$. Explain the error and give the correct answer. **[Find the error]**

14. The area of a square is $A = s^2$. If $s = 2x$, write A in expanded form. **[Area (substitute)]**
15. A rectangle has length $(3x + 2)$ and width $(x - 1)$. Find its perimeter in simplified form. **[Perimeter (polynomial)]**
16. Simplify: $3(2x - 1) - 2(x - 4)$. **[Expand & simplify]**

17. Simplify: $(10x^3 - 15x^2 + 5x) \div 5x$. **[Divide polynomial]**

18. Simplify: $(2a^3b^2)^2$. **[Power of a monomial]**